

Automated Water Service Area Delineation for Statewide GIS Modeling

An ArcGIS Python Toolbox that merges EPA public water system data into a clean, scope-aware water service boundary for Ohio parcels

The Problem

A statewide land use model needed to know, for every parcel in Ohio, whether it sits inside a public water service area. That drives the with-water versus without-water classification downstream. The source data for that comes from the EPA's Community Water System (CWS) service area dataset, but turning it into something usable for a single state runs into a handful of real obstacles.

- The EPA dataset ships as a large GeoPackage inside a zip hosted on GitHub, containing every state's data, not just Ohio's. Pulling only what's relevant means filtering by the right field on the right layer, not just clipping geographically.
- Community water systems aren't the whole picture. The EPA dataset also separately tracks transient and non-transient non-community systems (T_NTNC), which need to be pulled in and merged, not ignored.
- Some projects also have a locally supplied private or local sewer boundary that needs to be folded into the same output, but only when one is actually provided.
- Service area boundaries from different sources and different systems overlap each other in places, and duplicate or overlapping polygons would double-count or misrepresent coverage in downstream processing.
- Re-downloading a large GeoPackage on every run wastes time once the data has already been retrieved once.

My Approach

I built this as a custom **ArcGIS Python Toolbox (.pyt)**, consistent with the rest of this pipeline, so it runs as a native ArcGIS Pro geoprocessing tool.

One-time download, then cache. The tool checks for a locally cached copy of the EPA GeoPackage first. Only if it's missing does it download the zip from GitHub, extract just the GeoPackage from inside it, and discard the rest, so every subsequent run skips the download entirely.

Filter at the source, not after the fact. Rather than pulling every state's data and clipping it down, the tool filters the CWS and T_NTNC layers directly by their Ohio-specific attribute fields (`(Primacy_Agency = 'OH')` and `(PRIMACY_AGENCY_CODE = 'OH')`) at read time, so only relevant records ever get processed.

Layer merging with an optional local input. The Ohio-filtered CWS and T_NTNC layers are merged together, and if the user supplies a private or local sewer boundary shapefile, that gets folded into the same merge. If no private layer is provided, the tool proceeds cleanly with just the two EPA layers rather than requiring it.

Scope-driven clipping. Like the rest of the pipeline, the tool builds its clip boundary from the project's TAZ polygons for whichever scope is selected (statewide, an ODOT district, a single county, or a multi-county selection), then clips the merged water layer to that boundary before any further processing.

Dissolve to resolve overlap. After clipping, all overlapping boundaries are dissolved into a single clean layer, so the same physical coverage isn't represented by multiple stacked polygons.

A simple, explicit output flag. Every polygon in the final output gets `Has_Water = 1`. Downstream, any parcel that doesn't spatially intersect one of these polygons is treated as `Has_Water = 0`. Keeping the flag logic this explicit avoids ambiguity about what the output actually represents.

Self-checking QA. The tool checks its own output for red flags, such as zero features after clipping, zero records from either EPA layer, or a suspiciously small output polygon count, and raises clear warnings rather than staying silent.

Technical Highlights

- **Language/Platform:** Python, ArcGIS Pro (`arcpy`) Python Toolbox
- **Data source:** EPA Community Water System Service Area Boundaries (v3.0), via the EPA ORD GitHub repository, delivered as a GeoPackage
- **Networking:** Streamed download with progress logging, SSL handling for corporate network compatibility, local caching to avoid redundant downloads
- **Geoprocessing:** Attribute-filtered layer reads from a GeoPackage, feature merge (including an optional user-supplied layer), clip to a dissolved TAZ boundary, dissolve to resolve overlap, field calculation, feature class export to geodatabase
- **Scope flexibility:** Statewide, ODOT district, county, and multi-county processing from a single tool
- **Reporting:** A QAQC text log covering run metadata, data source and caching status, per-layer record counts at every stage (CWS, T_NTNC, private sewer, merged, clipped, dissolved, output), and any QA flags raised

Why It Matters

Whether a parcel has water service access is a direct input into how it gets classified in the larger land use model, so an incomplete or double-counted water layer would quietly distort that classification for however many parcels it touches. Filtering to Ohio at the source, merging in every relevant layer (including an optional local one), and dissolving away

overlap before the boundary reaches downstream steps means the water classification each parcel receives reflects the actual data, not an artifact of how it was assembled.